DANIEL RUIZ FIGUEROA

COMPUTER ENGINEER

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PROFILE

Computer Engineer based in Madrid, working as a Calypso developer in capital markets and heading for the space sector. I built SKYFALL, an open-source tool that maps where re-entering space debris is likely to fall, and a spacecraft descent and landing simulator as my final degree project.

EXPERIENCE

Calypso Developer · Qaracter · job title: Financial Analyst

Jan 2025 – Present · Madrid

- Develop and maintain trade confirmations for FX products on the Calypso platform.
- Generate the confirmation documents with the specific data each product requires.
- Manage delivery across confirmation channels: email, fax, TradeServ and SWIFT.

Calypso · Java · Oracle SQL

PROJECTS

SKYFALL · Re-entry risk maps for space debris · open source (MIT)

2026

- Turns official re-entry warnings (TIP messages from Space-Track) into 90, 95 and 99 % probability zones on a 3D globe, and compares them with the airspace closure strips from the EASA bulletins.
- Samples re-entry times from the measured error of a year of past warnings (about 4,450 warnings, 680 objects), propagates each orbit with SGP4 and scatters debris with Monte Carlo runs on a 0.5° grid.
- Validated on 100 real warnings the model had never seen: the 95 % zone contained 95 confirmed impact points, against 87 for the 120 km strip, for about the same closed area.

Python · FastAPI · pytest · React · TypeScript · CesiumJS · developed with Claude Code

Spacecraft Descent and Landing Simulator · Final degree project · solo · graded 8.9 / 10

2024

- Simulates descent and landing on the Moon, Mars and a configurable planet: gravity, wind, thrust, fuel burn that lightens the vehicle, hull heating, heat shield and parachute.
- Mission parameters are set before each run; vehicle models are streamed at runtime from Microsoft Azure Blob Storage with private access keys, with a local fallback.

C# · Unity · Azure Blob Storage · REST

Love Engine · Custom 3D engine in C++ · team project

University

- Engine built on Ogre3D (rendering), Bullet (physics), FMOD (audio) and Lua scripting, with a complete 3D title developed on top of it.

C++ · Lua · Ogre3D · Bullet · FMOD

EDUCATION

BSc in Video Game Development · Universidad Complutense de Madrid

2019 – 2024

- Programming in C++ and C#, physics simulation, artificial intelligence and machine learning, networking, and team projects run with Scrum and QA.

SKILLS

LANGUAGES	Python, C#, C++, C, Java, TypeScript, JavaScript, SQL
SPACE & SIMULATION	SGP4 orbit propagation, Monte Carlo methods, rigid-body physics, Unity, CesiumJS
BACKEND & CLOUD	FastAPI, REST APIs, Microsoft Azure, Oracle SQL, SQLite
FINANCIAL SYSTEMS	Calypso, FX trade confirmations, SWIFT
TOOLS & PRACTICE	Git, pytest, unit testing, Visual Studio, VS Code, Jupyter, Scrum

LANGUAGES

SPANISH	Native
ENGLISH	B2 · Cambridge First Certificate

OTHER

DRIVING LICENCE	B